

Gantt Chart Practice

Saturday, July 18, 2026 9:49 PM

HFN Class

P _i	B.T.	A.T.	I/O	Turn.
P ₁	12 16 8 4 7 10 5 2 6 3 0	0	4s CPU → 2s I/O 2s " → 4s "	6
P ₂	2 15 10 8 3 0	1	5s CPU → 1s I/O	1
P ₃	18 16 12 9 6	4	NA	
P ₄	7 5 25 0 9 2 0	20	4s CPU → 5s I/O	

Gantt Chart

0	2	4	6	8	11	13	15	17	22	24
P ₁	P ₂	P ₃	P ₁	P ₂	P ₁	P ₂	P ₃	P ₁	P ₄	
P ₂	P ₄	P ₁	P ₁	P ₃	P ₄	P ₄	P ₃			
24	25	27	29	30	38	40	41	45		
I/O Queue										
0	8	10	11	12	22	26	27	32		
~~~~~	P ₁	~~~~~	P ₂	~~~~~	P ₁	~~~~~	P ₄			

Q₁ (RR-2)

0 - P₁  
2 - P₂  
4 - P₃  
6 - [ ]  
10 - P₁  
11 - P₁  
13 - P₂  
15 - [ ]  
20 - P₄  
22 - P₄  
24 - [ ]  
26 - P₁  
27 - P₁  
29 - [ ]  
32 - P₄  
38 - P₄  
40 - [ ]

Q₂ (RR-4)

2 - P₁  
4 - P₁, P₂  
6 - P₁, P₂, P₃  
8 - P₂, P₃  
11 - P₃  
13 - P₃, P₁  
15 - P₃, P₁, P₂  
19 - P₁, P₂  
22 - P₂  
24 - P₂, P₄  
25 - P₄  
27 - [ ]  
29 - P₁  
30 - [ ]  
40 - P₄  
41 - [ ]

Q₃ (RR-8)

19 - P₃  
30 - P₃  
38 - P₃  
41 - P₃  
45 - [ ]

I/O Queue

8 - P₁  
10 - [ ]  
11 - P₂  
12 - [ ]  
22 - P₁  
26 - [ ]  
27 - P₄  
32 - [ ]

TurnA	Wait	Res
P ₁ -30	12	0
P ₂ -24	15	1
P ₃ -41	23	0
P ₄ -21	9	2
Avg → 29	14.75	0.25

3. Given the arrival time and burst time for the processes below, draw a gantt chart using multilevel feedback queue scheduling algorithm showing the states of the ready queues of different levels. Then calculate the average waiting time, response time and turnaround time. There are 3 queues with decreasing priority all using round robin scheduling algorithm internally. **Q1 (quantum: 3, priority:0), Q2 (quantum:5, priority: 1), Q3 (quantum:10, priority:2).** In case of I/O request, a process will be promoted one

Process	Arrival Time	Burst Time	I/O Request
P1	0	10	3s IO after 7s CPU
P2	2	23	4s I/O after 5s CPU 3s I/O after 12s CPU
P3	5	18	N/A
P4	7	16	2s I/O after 11s CPU

level upon I/O completion.

**Answer:**

Avg. Waiting: 27.25

Avg. Response: 1

Avg. Turnaround: 47.25

Proc.	Burst Time	A.T.	I/O Queue	Term
P ₁	10 7	0	7s CPU → 3s IO	
P ₂	23 5 12	2	5s CPU → 4s IO 12s CPU → 3s IO	
P ₃	18	5	N/A	
P ₄	16 11	7	11s CPU → 2s IO	

Gantt Chart


I/O Queue

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Q₁ (RR-3)

Q₂ (RR-5)

Q₃ (RR-10)

I/O Queue

**TunnA**

Wait

Res

P-

 $\rho_2 -$ 

B-

$P_4 -$

Proc.	Burst Time	A.T.	I/O Queue	Term
$P_1$			s CPU $\rightarrow$ s IO	
$P_2$			s CPU $\rightarrow$ s IO	
$P_3$			s CPU $\rightarrow$ s IO	
$P_4$			s CPU $\rightarrow$ s IO	


## I/O Queue

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Q₁ (RR-2)

Q₂ (RR-4)

Q₃ (RR-8)

I/O Queue

TurnA

Wait

Res

P₁-

P₂-

P₃-

P₄-

Tuesday, July 21, 2026 1:45 AM

# Gradient Boosting

Tuesday, July 21, 2026

1:45 AM